

1 non-reg $\cap$ non-reg = reg

Lets use the following non-regular languages A and B :

$$A = \{0^n 1^n \mid n \geq 0\}$$

$$B = \{2^n 3^n \mid n \geq 0\}$$

These languages are irregular for the same reason, they are not closed under pumping lemma.

Lets suppose that A is regular, let P denote pumping length and $s = 0^P 1^P$. According to pumping lemma s splits into XY^iZ so that rules i -iii) of the pumping lemma hold.

Now $Y = 0^j \mid j \in \{1 \dots P\}$ and if we pump down and set $i = 0$ we get $s' = XZ = 0^{P-j} 1^P$ which is not in A .

Since these languages do not use the same alphabet, they are not equal and the only intersection they have, the empty string, would be regular

$$A \cap B = \{\emptyset\}$$

2 File trees

Lets define the language DIR we have two major rules

1. Each file, a , has a corresponding pair of parenthesis
2. The root folder does not have a file name

Lets show that this language is irregular by applying pumping lemma.

Lets suppose DIR is regular, let P denote pumping length and $s = (a^{P-2}())$. According to pumping lemma s splits into XY^iZ so that rules i -iii) of the pumping lemma hold.

Now $Y = a^j$ where $(j \mid j \in \{1 \dots P\})$ and if we pump down and set $i = 0$ we get $s' = XZ = a^{P-j} (a^{P+1})^{P+1}$ and since $P + 1 \neq (P - 1) + 1$ which is not in does not follow the rules of DIR therefore DIR is not regular.

3 CRC codes

3.a

The checksum of the empty string would have a draw between all the numbers: $\{0, 1, 2\}$ and since 0 takes precedence, the checksum would be $[0, 0]$.

3.b

Suppose that the following language is regular:

$$A = \{0^i 1^j 2^k \mid \text{file has valid checksum}\}$$

- Let P denote pumping length
- $s = 0^P 1^{P+1} 2^{P-1}$

According pumping lemma s splits into XYZ so that rules *i-iii*) of the pumping lemma hold.

According to rules *ii*) and *iii*) of the pumping lemma $|xy| \leq P$ this means that

$$y = 0^j \mid j \in \{1 \dots P\}$$

From rule *i*), if we pump down and set $i = 0$ we get $s' = XZ = 0^{P-j} 1^{P+1} 2^{P-1}$

Because the checksum takes the lowest number in case of a draw, after pumping the checksum will be invalid and that is a contradiction. Therefore the language is not regular.